Araz Mehrabani
Mechanical Engineer | Wind Loads · Aeroelasticity · OpenFAST

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PORTFOLIO
Technical Project Examples
Google Drive

PROFILE & CORE COMPETENCIES

Mechanical engineer focused on **aeroelastic load calculation and wind-turbine dynamics.** Hands-on experience with OpenFAST model setup, IEC/DNV DLC campaigns, component loads, fatigue/extreme analysis, and Python-based simulation/data workflows.

TECHNICAL FOCUS

Aeroelastic Loads: OpenFAST; DLC 1.2/1.3/2.3/5.1/6.1/7.1/11.1; IEC 61400-1, DNV-ST-0437; multi-seed campaigns.
Fatigue & Extremes: MLife, MExtremes, rainflow counting, DEL, Miner's rule, load collectives and max/min evaluation.
Python & Data: NumPy, pandas, Matplotlib, SciPy; input editing, case/folder generation, batch execution, .out parsing, CSV/Excel export.
Component Loads: blade root, main shaft, yaw bearing, tower sections/base, generator signals, tip clearance; time-series/channel comparison.

PROFESSIONAL EXPERIENCE

Development Engineer — Sabaniroo, Tehran **01/2022–02/2023**

- Set up/adapted **OpenFAST** models and multi-seed DLC campaigns to **DNV-ST-0437**; extracted and compared component channels and evaluated fatigue/extreme loads using MLife/MExtremes.
- Built a **Python** workflow for input files, case/folder generation, batch execution, .out parsing, data export and plotting across several hundred simulations; approx. **80% less manual effort**.

Mechanical Engineer — TehranRigzar, Tehran **03/2017–12/2021**

- Built simplified **ANSYS** models of a vibrating screen with local mesh refinement; evaluated stresses, natural frequencies and mode shapes to avoid resonance, supported by experimental spring testing.

Working Student – Mechanical Design — Ario Fidar Maham Co., Tehran **11/2012–05/2013**

- Recreated a 40x90 jaw crusher in **CATIA V5**, including individual components, assemblies and manufacturing drawings.

WIND LOADS & RESEARCH

Master's Thesis (ongoing; expected 11/2026) – Rotor as a Sensor — Python/OpenFAST pipeline using Coleman/Feingold transformation for 0P/1P/2P blade-load harmonics, filter comparison and FFT/order diagnostics; yaw-misalignment sensitivity at -8/0/+8 deg.

20 MW Offshore Wind Turbine – Loads & Dynamics — Adapted the IEA 15 MW reference to a 20 MW monopile concept; several hundred DLC simulations, component-channel evaluation, rainflow/DEL/Miner and MLife/MExtremes processing.

EDUCATION

M.Sc. Wind Energy Engineering, Hochschule Flensburg — current grade: 1.9 **03/2023–11/2026**
M.Sc. Mechanical Engineering, Hakim Sabzevari University — Control & Robotics **09/2014–01/2017**
B.Sc. Mechanical Engineering, Islamic Azad University — Structural Dynamics **09/2008–06/2013**

LANGUAGES

German: B2 (actively improving). English: C1 / IELTS 7.0. Persian: Native.